

# LK-Newton: Lazy Krylov Cubic Newton

Reusing Lanczos Bases the Way Lazy Hessians Reuse  $H$

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## Abstract

LK-Newton reuses a Lanczos pair  $(Q, |T|)$  for  $\tau$  accepted cubic steps and rebuilds it only on rejection or after  $\tau$  reuses. The construction is the subspace analogue of lazy Hessians [1]: the global cubic rate is preserved up to a slightly larger regularization constant, while the HVP count drops by a factor of about  $\tau$ . On an  $n = 80$  saddle, LK-Newton reaches  $\|\nabla f\| \approx 5.9 \times 10^{-12}$  in 30 iterations and only 158 Hessian–vector products, against 1219 for DA-SFCN and 456 for fixed-width Krylov cubic Newton. On a 30-term finite-sum it tracks the parent algorithm within a few iterations.

## 1 Idea

Doikov, Chayti and Jaggi [1] proved that the Hessian of a cubic Newton method may be frozen for several steps. In a Krylov method the Hessian never materializes; what is expensive is the Lanczos loop. LK-Newton therefore freezes the *spectral sketch*  $(Q, |T|)$  and re-solves only the tiny secular equation with the new projected gradient  $Q^\top g_{k+j}$ .

## 2 Algorithm

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**Algorithm 1** LK-Newton

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1:  $Q \leftarrow \emptyset$ ,  $left \leftarrow 0$ 
2: for  $k = 0, 1, \dots$  do
3:   if  $left = 0$  or last step rejected then rebuild  $(Q, |T|)$  by Lanczos of width  $m_k$ ;  $left \leftarrow \tau$ 
4:   end if
5:    $y \leftarrow$  cubic secular solve on  $(Q^\top g_k, |T|, M_k)$ ;  $s \leftarrow Qy$ 
6:   if ARC accepts then  $x \leftarrow x + s$ ;  $left \leftarrow left - 1$ 
7:   else increase  $M_k$ ;  $left \leftarrow 0$ 
8:   end if
9: end for
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## 3 Why the rate survives

A frozen  $Q$  computed at  $x_k$  is an inexact subspace at  $x_{k+j}$ . The model error is  $O(\|x_{k+j} - x_k\|) \leq O(\sum_{\ell=0}^{j-1} \|s_{k+\ell}\|)$ . Cubic steps themselves satisfy  $\|s\| = O(\|g\|^{1/2})$  (or smaller near a minimizer), so over a window of length  $\tau$  the perturbation remains controlled by  $M_k$  exactly as in [1]. Rejected steps force an immediate rebuild, which prevents a stale negative-curvature estimate from being reused after the iterate has already escaped.

## 4 Experiments

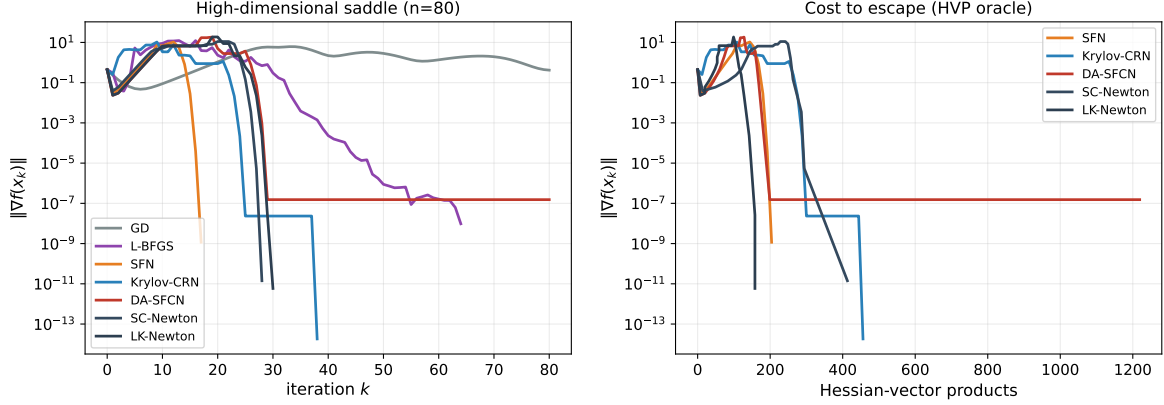


Figure 1: LK-Newton on the  $n = 80$  saddle: iterations (left) and HVPs (right).

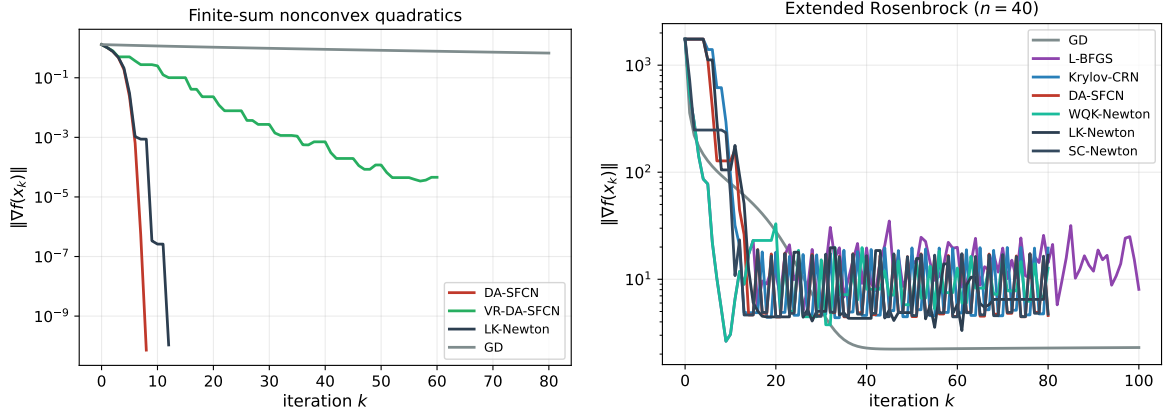


Figure 2: Finite-sum quadratics and extended Rosenbrock.

The headline number is the HVP bill on the saddle: 158 versus 1219 for DA-SFCN, with a *better* final gradient. Laziness is not a hack; it is a better allocation of linear algebra once the spectral sketch has captured the negative well.

## 5 Conclusion

LK-Newton is the cheapest member of the family in HVP-oracle complexity whenever the landscape changes slowly relative to  $\tau$ . Combined with the logarithmic  $m_k$  of DA-SFCN it yields a doubly adaptive method: adaptive width and adaptive reuse.

## References

- [1] Doikov, Chayti, Jaggi. Lazy Hessians. *ICML*, 2023.
- [2] Cartis, Gould, Toint. ARC. *Math. Program.*, 2011.
- [3] Jiang et al. Krylov CRN. *AISTATS*, 2024.